1️⃣ Bubble Sort

Logic: Adjacent elements compare karo aur swap karo.

Memory Trick

n passes
har pass me adjacent compare
bada element end me push

Steps

1. Outer loop → n-1
2. Inner loop → adjacent compare
3. If arr[j] > arr[j+1] → swap

Example

5 3 8 2

3 5 8 2
3 5 2 8
3 2 5 8
2 3 5 8

👉 Largest element **bubble ho ke end pe chala jata hai**

2️⃣ Insertion Sort

Logic: Card game jaisa.

Memory Trick

Left side sorted
Right side element insert karo

Steps

1. key = arr[i]
2. previous elements shift
3. correct position pe insert

Example

8 3 5 2

3 8
3 5 8
2 3 5 8

3️⃣ Selection Sort

Logic: Har pass me **minimum element find karo**.

Memory Trick

min find → swap with first unsorted

Steps

1. min = i
2. smallest find
3. swap with arr[i]

Example

7 4 9 1

1 4 9 7
1 4 9 7
1 4 7 9

4️⃣ Linear Search

Logic: Ek ek element check karo.

Memory Trick

for loop
if arr[i] == key
return index

Example

2 4 6 8
search = 6

Check

2 → no
4 → no
6 → yes

5️⃣ Binary Search (Iterative)

Logic: Half half search.

Memory Trick

mid = (low+high)/2

Steps

1. compare with mid
2. key < mid → left
3. key > mid → right

Example

1 3 5 7 9
search 7

mid=5
7>5 → right
mid=7 → found

6️⃣ Binary Search (Recursive)

Same logic but function repeat hota hai.

Memory trick

if key < mid → left recursion
if key > mid → right recursion

7️⃣ Quick Sort

Logic: Pivot choose karo.

Memory Trick

pivot
left < pivot
right > pivot

Steps

1. choose pivot
2. partition
3. recursively sort left & right

Example

8 3 5 2
pivot = 8

3 5 2 | 8

8️⃣ Merge Sort

Logic: Divide and merge.

Memory Trick

divide
divide
merge sorted arrays

Example

8 3 5 2

Divide

8 3 | 5 2

Sort

3 8 | 2 5

Merge

2 3 5 8

9️⃣ Activity Selection

Logic: Maximum activities without overlap.

Memory Trick

sort by finish time
select first activity
next start ≥ previous end

Example

(1,2)
(3,4)
(0,6)
(5,7)
(8,9)

Selected

(1,2)
(3,4)
(5,7)
(8,9)

🔟 Fractional Knapsack

Logic: Profit/Weight ratio.

Memory Trick

ratio = profit/weight
sort descending
take full item
take fraction if needed

Example

Weight Profit
10 60
20 100
30 120

Ratio

6 5 4

Take items with highest ratio.

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11 Prim's Algorithm

Logic: Minimum Spanning Tree.

Memory Trick

start from any node
choose minimum edge
connect new vertex
repeat

Example

Edges

1-2 =2
2-4 =3
1-3 =6

Total = 11

---

12 Job Scheduling

Logic: Maximum profit jobs.

Memory Trick

sort by profit
schedule before deadline
latest free slot

Example

Job Profit Deadline
J1 100 2

J2 19 1
J3 27 2

Result

J3 J1

🎯 Super Easy Way to Remember All 12

Type	Algorithms
Sorting	Bubble, Insertion, Selection
Searching	Linear, Binary
Divide & Conquer	Quick, Merge
Greedy	Activity, Knapsack
Graph	Prim
Scheduling	Job Scheduling

✅ Exam Tip

Agar tum ye **4 core ideas yaad rakh lo**, saare algorithms yaad rahenge:

Swap → Bubble
Insert → Insertion
Divide → Merge
Pivot → Quick
Ratio → Knapsack
Finish time → Activity
Minimum edge → Prim
Profit → Job Scheduling